Residuals Matter, Noise Still Wins: Insights into Plant-Soil Segmentation

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***Abstract*—We compared traditional segmentation methods, classical machine learning baselines, and deep learning models for plant-soil segmentation on the EWS dataset. The methods include ExG, HSV thresholding, K-means, edge-based segmentation, Watershed, GrabCut, DenseCRF, Logistic Regression, Random Forest, U-Net, ResU-Net, ASPP-ResU-Net, and Attention-Gated ASPP-ResU-Net. Among the traditional methods, simple colour-based approaches such as ExG (Plant IoU 0.754) and HSV thresholding (0.708) outperformed more complex alternatives such as GrabCut and Watershed, showing that explicit colour priors remain surprisingly competitive. All deep models surpassed the Plant IoU benchmark of 0.775 reported in the original paper, with the best model reaching 0.8327. The largest clean-set improvement appeared when moving from U-Net to ResU-Net, while ASPP and attention gates provided smaller additional gains. However, the robustness results revealed a more nuanced picture. Under strong Gaussian noise, all residual-based models collapsed near zero Plant IoU, while the plain U-Net retained a Plant IoU of 0.27 at the highest severity—a complete ranking reversal relative to clean-set performance. This suggests a meaningful accuracy-robustness trade-off: the residual-based designs that improve clean-set accuracy in our setting also appear substantially more vulnerable to severe Gaussian noise.**

***Index Terms*—Plant-soil segmentation, semantic segmentation, traditional segmentation, deep learning, robustness, U-Net, residual learning, agricultural vision.**

I. Introduction

Plant-soil segmentation is an important task in precision agriculture because it supports later analysis such as crop monitoring, field measurement, and automated decision-making. In real field scenes, however, this task is not as simple as it first appears. Compared with controlled laboratory images, field images are much harder to handle because lighting changes constantly, soil appearance is often uneven, and environmental conditions differ across time and location. As a result, even binary segmentation can become difficult in practice.

In this project, we use the EWS Dataset released through the ETH Research Collection [1]. The dataset contains RGB field images together with pixel-level masks for plant-soil segmentation. This provides a useful setting for comparing different methods under the same conditions. More importantly, it allows us to examine how these methods behave not only when the images are clean, but also when the visual conditions become less ideal.

Based on this setting, the main question of the project is straightforward: which kinds of methods are most effective for plant-soil segmentation in realistic field conditions? To answer this question, we compare both classical baselines and deep learning models within a unified experimental pipeline. We evaluate them using the same preprocessing and metrics, and we consider both clean-set performance and robustness under corruption. In this way, the project is not only about identifying the best-performing model, but also about understanding where the improvement actually comes from.

II. Literature Review

Earlier segmentation methods in agricultural vision often relied on handcrafted rules. A common example is the use of vegetation indices such as Excess Green (ExG), sometimes combined with thresholding or simple classifiers. These methods are still useful because they are easy to implement, interpretable, and computationally cheap. However, they usually depend heavily on stable colour contrast between vegetation and background. Once illumination changes or the soil becomes visually complex, their performance can drop quickly. For this reason, they are helpful as baselines, but they are usually not strong enough on their own for more difficult field conditions.

With the development of deep learning, segmentation methods gradually shifted toward representation-based models. Encoder-decoder architectures, especially U-Net and its variants, became widely used because they can learn pixel-level segmentation directly from data while still preserving spatial detail through skip connections. This is important in agricultural images, where plant regions may be small, irregular, or visually close to the background. In these cases, the model needs more than colour information alone; it also needs to use local texture, boundary cues, and surrounding context.

Later studies extended this general framework in several ways. Residual connections were introduced to improve optimization and feature reuse, especially in deeper networks. Multi-scale modules such as ASPP were designed to capture information at different receptive-field sizes, while attention mechanisms were used to guide the model toward more relevant regions and suppress background interference. Although these ideas are different in form, they are all trying to solve

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Contributions. J. Bai maintained the shared codebase and the `ImagePipeline` abstraction, and implemented and evaluated the deep-learning models (U-Net, ResU-Net, ASPP-ResU-Net, AG-ASPP-ResU-Net) together with the robustness protocol. C. Chen implemented the classical machine-learning baselines (Logistic Regression, Random Forest) and their feature-configuration study. J. Liu worked on the traditional computer-vision methods (ExG, HSV, K-means, edge-based, Watershed, GrabCut, DenseCRF). A. Zhang led the writing of this report. X. Dong prepared the accompanying presentation slides.

a similar problem: how to make segmentation more accurate and more reliable in visually complicated scenes.

Recent work in plant phenotyping also shows that model performance depends strongly on real-world variability. Deep learning methods have already shown strong results for outdoor plant segmentation under diverse field conditions [2]. More recent studies have further explored how to improve crop segmentation through stronger supervision and additional geometric information, such as depth-informed pseudo-labelling and large-scale depth-labelled data [3], [4]. These studies suggest that high clean-set accuracy is important, but robustness under realistic variation is just as important.

Following this line of work, our project adopts a progression from simple baselines to more expressive deep models. This design is useful because it allows us to compare not only which model performs best, but also why the performance changes. In other words, we are interested in whether the gain mainly comes from handcrafted colour knowledge, stronger nonlinear classification, or better learned spatial representations.

III. Methods

We treat the task as binary semantic segmentation, where each pixel in a field image is labelled as plant or soil. This pixel-level setting is appropriate for agricultural applications that need accurate estimates of plant coverage and boundaries [5]. All images and masks are resized to 352×352 so that every method is tested under the same conditions.

Our goal is not only to rank a set of models, but to understand why some of them work better than others. For that reason, the experimental design moves from traditional segmentation methods to machine learning baselines and then to deep learning models. This progression helps reveal whether the main improvement comes from handcrafted colour priors, stronger nonlinear classification, or learned spatial representations.

A. Traditional Segmentation Methods

We begin with traditional, non-learning segmentation methods because they represent the simplest way to approach the problem and set the floor for what explicit colour and geometry rules can achieve without any training data.

The most direct baseline is **Excess Green (ExG)**, computed as $2G - R - B$. ExG is widely used as a zero-training vegetation prior because it emphasizes green-dominant regions in RGB imagery [6]. We include it both as a standalone baseline and as an auxiliary feature in later ablations.

HSV thresholding provides a colour-space alternative by isolating pixels whose hue falls within the green band, separating chromatic content from brightness. This makes it potentially more robust than ExG under illumination changes [8].

K-means clustering (k=2) segments the image by grouping pixels in RGB colour space without supervision. The cluster with the higher ExG value is assigned as plant, removing dependence on a fixed colour threshold.

Edge-based segmentation applies Gaussian blur followed by Canny edge detection and morphological closing. This tests whether plant-soil boundaries can be reliably found from low-level gradients alone.

Watershed uses ExG-derived seed markers (definite plant: ExG > 20; definite soil: ExG < -20) to drive region-based separation via graph-cut flooding, allowing spatial context to resolve ambiguous boundary regions.

GrabCut initialises foreground and background Gaussian mixture models from ExG labels and refines them iteratively over five EM steps, offering a more expressive probabilistic alternative to Watershed.

DenseCRF uses ExG as a unary prior (gt_prob=0.7) and refines predictions with pairwise Gaussian (s_{xy}=3) and bilateral (s_{xy}=60, s_{rgb}=13) potentials over five inference iterations. It serves as a probabilistic post-processing baseline that enforces spatial smoothness without requiring a learned classifier.

B. Classical Machine Learning Baselines

Building on the traditional baselines above, the classical machine learning methods introduce pixel-wise supervision on top of handcrafted features. This step asks whether learning a decision rule from labelled pixels, rather than thresholding a colour index directly, can extend the ceiling of what handcrafted features alone can deliver.

Logistic Regression is the first learned pixel-wise baseline. Unlike ExG, which relies on a fixed colour rule, Logistic Regression learns a linear decision boundary from handcrafted features [7]. Each pixel is treated as a sample, and we test three feature sets: RGB, RGB+HSV, and RGB+HSV+ExG. HSV is included because it separates chromatic content from brightness and may help when illumination is less stable [8]. The model is trained with `max_iter=1000`, balanced class sampling, and a fixed random seed of 42. This gives us a simple but informative reference for what can be achieved through feature engineering and linear classification alone.

We then introduce Random Forest as a stronger nonlinear baseline. Compared with Logistic Regression, Random Forest can capture more complex interactions among handcrafted features because it combines an ensemble of decision trees [9]. To keep the comparison consistent, we use the same three feature settings—RGB, RGB+HSV, and RGB+HSV+ExG—and train the model with `n_estimators=100`, `max_depth=15`, `samples_per_class=2000`, and `random_state=42`. This allows us to see how much is gained by replacing a linear classifier with a more flexible nonlinear one, while still staying within the handcrafted-feature setting.

Together, the classical baselines define the point at which handcrafted colour information begins to reach its limit. They remain attractive because they are simple and interpretable, but their reliance on manually designed features also makes them sensitive to changing field conditions. This naturally leads to the next step of the study: testing whether deep models can move beyond that limitation by learning stronger spatial representations directly from the images.

C. Deep Learning Segmentation Models

Once the comparison moves past the classical baselines, the central question changes slightly. The issue is no longer

whether a handcrafted feature can separate plant from soil, but whether learned spatial representations can do so more reliably under realistic field variation. For this reason, the deep-learning part of the study focuses on encoder-decoder architectures, which are designed to combine local detail with broader contextual information.

The natural starting point for the deep-model comparison is **U-Net**. It provides a clear baseline because it already captures the essential encoder-decoder idea: semantic abstraction in the contracting path, spatial recovery in the expanding path, and skip connections that preserve fine detail across the network [10]. In this study, U-Net is not treated as a weak baseline, but as the model that defines the standard against which all later improvements need to justify themselves.

The first major refinement to the U-Net baseline is **ResU-Net**, which introduces residual learning into the encoder-decoder structure. This is a particularly meaningful step in the study because residual connections are not mainly about making the architecture more complicated; they are about making learning more stable and feature propagation more effective [11]. In our implementation, the residual blocks are paired with ReLU activations and Group Normalization to suit the small-batch training setting [14]. Conceptually, ResU-Net tests whether the main bottleneck of the baseline is poor optimization rather than insufficient architectural sophistication. This later becomes one of the clearest patterns in the results.

The next model, **ASPP-ResU-Net**, builds on the residual backbone by introducing **Atrous Spatial Pyramid Pooling (ASPP)**. ASPP is designed to aggregate information across multiple receptive fields, allowing the model to combine local structure with larger-scale context [12]. This is a meaningful extension in field imagery, where plant-soil separation may depend not only on local appearance but also on surrounding texture and regional context. In this sense, ASPP-ResU-Net asks whether multi-scale contextual modelling can further improve a backbone that is already strong in optimization and feature reuse.

The final model in the deep-learning comparison is **Attention-Gated ASPP-ResU-Net**, which adds attention-based skip refinement to the ASPP-ResU-Net architecture. Attention mechanisms help the model emphasize informative regions while suppressing less relevant background responses [13]. Here, attention gating is used to make skip fusion more selective, with the goal of improving foreground-background separation in cluttered field scenes. This model therefore represents the most refined architecture in the study: not simply deeper or larger, but more selective in how information is propagated through the network.

To conclude, the four deep architectures are not simply a list of alternatives. They represent a structured sequence of hypotheses: U-Net tests the baseline encoder-decoder paradigm; ResU-Net tests whether better optimization and feature reuse improve the task; ASPP-ResU-Net tests whether multi-scale context matters; and Attention-Gated ASPP-ResU-Net tests whether more selective feature fusion can refine the strongest model further. Table I summarises all implemented methods.

TABLE I
SUMMARY OF IMPLEMENTED METHODS

Method	Type	Key design
ExG	Traditional	Colour index, zero training
HSV thresholding	Traditional	Hue-range mask
K-means	Traditional	Unsupervised colour clustering
Edge-based	Traditional	Canny + morphological close
Watershed	Traditional	ExG-seeded region flooding
GrabCut	Traditional	Iterative GMM foreground/background
DenseCRF	Traditional	ExG unary + pairwise smoothing
Logistic Reg.	Classical ML	Linear, RGB/HSV/ExG
Random Forest	Classical ML	Ensemble, nonlinear
U-Net	Deep	Encoder-decoder, skip connections
ResU-Net	Deep	+ Residual blocks, GroupNorm
ASPP-ResU-Net	Deep	+ Multi-scale atrous pooling
AG-ASPP-ResU-Net	Deep	+ Attention-gated skip fusion

D. Training Configuration and Implementation Choices

All deep models are trained under the same protocol so that differences in performance can be attributed to architecture rather than inconsistent optimization. Input images are normalized to $[0, 1]$, and the segmentation masks are converted into binary form with plant pixels treated as the positive class.

The loss function combines **binary cross-entropy with logits** and **Dice loss** in equal proportion. This choice reflects the fact that the task requires both stable pixel-wise learning and strong overlap quality between prediction and ground truth. Binary cross-entropy supports optimization at the pixel level, while Dice loss directly emphasizes region-level agreement, which is particularly useful in segmentation settings with foreground-background imbalance.

Optimization is carried out using **AdamW** with an initial learning rate of 1×10^{-4}. Learning rate reduction on validation plateau and early stopping are both applied so that the comparison reflects the best achievable model state rather than arbitrary final-epoch behaviour. In the main architecture comparison, the batch size is set to 3, the maximum number of epochs is 100, and the inference threshold is 0.6. A fixed random seed of 42 is used throughout the experiments. During training, checkpoints are saved and the best-performing checkpoint on validation loss is used for final evaluation.

The framework also supports feature-augmented inputs, including ExG and HSV-derived channels. These are introduced only in ablation experiments, allowing the study to test whether explicit agricultural priors still provide useful information once the model is already able to learn hierarchical representations directly from RGB input.

E. Evaluation Design and Robustness Protocol

Our evaluation is designed to do more than identify the highest clean-set score. All methods are first tested on the standard train/validation/test split, using the same preprocessing pipeline and the same resized image resolution. Performance is measured with class-wise IoU, precision, recall, F1-score, overall accuracy, confusion matrices, and inference time.

A second part of the evaluation focuses on robustness. Since field imagery is rarely collected under ideal conditions, the models are further tested under several controlled corruption types: **Gaussian noise**, **Gaussian blur**, **brightness shift**, **affine rotation**, and **JPEG compression**. Each corruption is applied at multiple severity levels so that degradation trends can be observed rather than inferred from a single perturbed setting.

This robustness-oriented design is central to the study. It allows the comparison to distinguish between models that perform well only under clean benchmark conditions and models that remain reliable when image quality changes. In this way, the evaluation framework supports a more practical interpretation of segmentation performance, one that is closer to real deployment conditions in agricultural environments.

IV. Experimental Results

A. Evaluation Protocol

All methods were evaluated under the same train/validation/test split and preprocessing pipeline described in Section III. We report class-wise IoU and F1-score, pixel accuracy, confusion matrices, and inference time. To examine reliability beyond the clean setting, robustness was further evaluated under controlled corruptions, including Gaussian noise, Gaussian blur, brightness shift, affine rotation, and JPEG compression, each tested at multiple severity levels. Gaussian noise was evaluated at five severity levels, with the highest level corresponding to variance 0.2.

B. Traditional Segmentation Methods

The traditional segmentation results show a clear and somewhat counterintuitive pattern: the simplest colour-based approaches outperform more elaborate region-based and probabilistic methods by a substantial margin. ExG achieves the strongest result in this group, with a Plant IoU of 0.754 and pixel accuracy of 0.903—numbers that already approach the classical machine learning baselines. HSV thresholding is the next strongest, with Plant IoU 0.708 and accuracy 0.897. Both methods require no training data and run in under 25 ms per image.

The more complex traditional methods do not improve on these simple baselines. DenseCRF, which refines ExG predictions with bilateral and Gaussian pairwise potentials, reaches Plant IoU 0.544—below ExG despite taking 3.1 s per image. GrabCut and Watershed produce weaker results still (Plant IoU 0.499 and 0.376 respectively). Watershed's low Soil IoU (0.211) reflects a structural limitation: when the soil region does not contain enough pixels with strongly negative ExG to seed definite-soil markers, Watershed floods large soil areas into the plant label. K-means and edge-based segmentation perform weakest, with Plant IoU below 0.30, confirming that unsupervised colour grouping and low-level gradient detection alone are insufficient for this task.

TABLE II
Test-set performance of traditional segmentation methods.

Method	Acc.	Plant IoU	Plant F1	Time (s)
ExG	0.9032	0.7536	0.8595	0.025
HSV thresholding	0.8972	0.7083	0.8292	0.011
DenseCRF	0.7494	0.5444	0.7050	3.114
GrabCut	0.6814	0.4992	0.6659	4.146
Watershed	0.4650	0.3758	0.5463	0.062
K-means	0.5831	0.2967	0.4576	1.560
Edge-based	0.5796	0.2614	0.4145	0.036

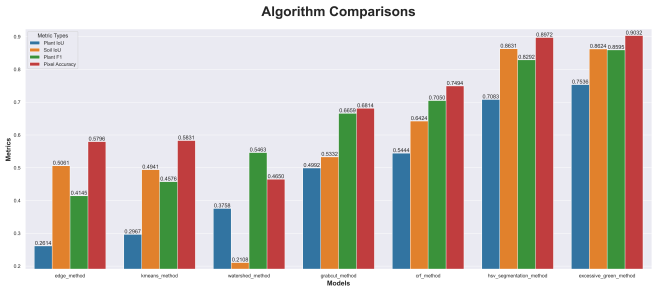


Fig. 1. Comparison of traditional segmentation methods on the test set. ExG and HSV thresholding achieve the strongest results despite requiring no training, while more complex methods such as DenseCRF and GrabCut underperform at substantially higher inference cost.

Table II summarises all traditional method results. Figure 1 compares them visually.

C. Classical ML Baselines and Feature Effects

The classical results already show a clear pattern. Plant-soil separation can be achieved to a reasonable extent using handcrafted colour information, but the quality remains limited once the visual conditions become less ideal. Both Logistic Regression and Random Forest improve when moving beyond RGB-only features, and the gain from adding HSV is especially clear. However, the extra ExG channel brings only a small improvement, particularly for Logistic Regression. This suggests that explicit feature engineering is useful, but only up to a point.

These results confirm that handcrafted features remain useful when the model itself is relatively simple. At the same time, they also show the ceiling of that approach. Even the best classical configuration still falls well short of the deep models, which suggests that the main challenge in this task is not only learning better colour thresholds, but also learning spatial structure more effectively. ExG is therefore best understood as a strong zero-training baseline rather than a competitive alternative to the deep models.

D. Deep Model Comparison on Clean Splits

The deep model results tell a more structured story than a simple ranking of performance. All four encoder-decoder

TABLE III
TEST-SET PERFORMANCE OF CLASSICAL METHODS UNDER DIFFERENT FEATURE CONFIGURATIONS.

Method	Feature set	Acc.	Plant IoU	Plant F1	Time (s)
Logistic Reg.	RGB	0.8689	0.7016	0.8247	0.048
Logistic Reg.	RGB+HSV	0.8774	0.7130	0.8325	0.098
Logistic Reg.	RGB+HSV+ExG	0.8774	0.7130	0.8325	0.104
Random Forest	RGB	0.8575	0.6665	0.7999	1.187
Random Forest	RGB+HSV	0.8597	0.6786	0.8085	1.144
Random Forest	RGB+HSV+ExG	0.8676	0.6902	0.8167	1.136

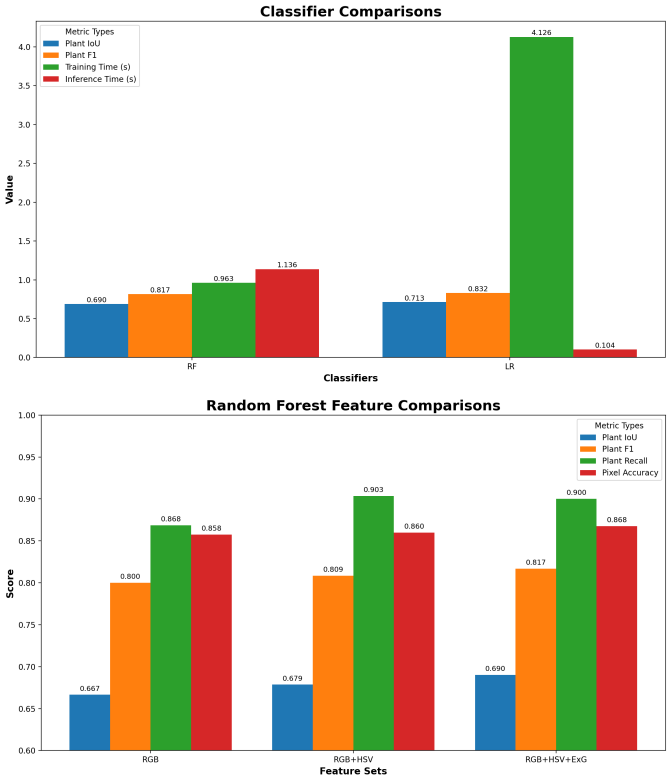


Fig. 2. Classical ML baselines. Top: Logistic Regression vs. Random Forest on Plant IoU/F1 and time. Bottom: Random Forest feature ablation—RGB→RGB+HSV gives the clearest gain, while adding ExG yields only a small further improvement.

architectures clearly outperform the classical baselines, and all of them surpass the best Plant IoU of 0.775 reported in the original paper [2], with our strongest model reaching 0.8327—an improvement of 5.77 percentage points on this metric. The most important change within the deep-model progression, however, happens early. Moving from U-Net to ResU-Net produces the largest improvement, while ASPP and attention gating provide smaller but still consistent refinements. The best clean-set performance is achieved by Attention-Gated ASPP-ResU-Net, although the gap between the top three deep models is relatively small compared with the jump from U-Net to ResU-Net.

A closer comparison shows that the largest single gain came from moving from U-Net to ResU-Net, with Plant IoU increasing by about 1.29 percentage points. By contrast, ASPP and attention gating produced gains of 0.03 and 0.06 percentage points respectively. This suggests that residual learning contributed more to final performance than later architectural refinements. The improvement from ASPP was especially limited, which is consistent with the nature of this task: in binary plant-soil segmentation, the scale variation is narrower than in more complex scene-segmentation problems, so the value of multi-scale receptive fields may be less pronounced.

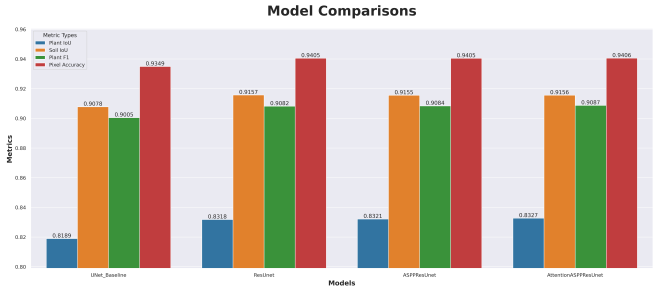


Fig. 3. Test-set metrics across the four deep architectures. The largest jump appears between U-Net and ResU-Net; ASPP and attention gating deliver smaller additional gains.

Attention gating did not change the ranking dramatically, but it appears to have improved the balance between precision and recall on the plant class, partly reducing the false-negative tendency seen in earlier models. The validation scores are also consistently slightly higher than the test scores. For the best model, the validation-test gap in Plant IoU is about 1.2 percentage points. This gap is attributable to a year-based distribution shift: the training and validation sets contain images from 2018–2020, while the test set contains images from 2017, where illumination and soil conditions differ. This is consistent with the year variability analysis in the original paper [2], and reflects a stable domain shift rather than unstable training.

E. Training Behaviour and Confusion-Matrix Evidence

The training curves show that all deep models converge in a stable way. In later epochs, training loss continues to decrease while validation loss tends to flatten or fluctuate slightly, which is consistent with mild overfitting and supports the use of early stopping and best-checkpoint selection. In our experiments, the best checkpoint often appeared around epoch 40, after which training loss kept decreasing but validation performance no longer improved. This pattern is typical of small-dataset settings, where overfitting becomes visible before the nominal end of training.

The confusion matrices tell a similar story from a different angle. The strongest models separate the two classes reliably, but the remaining mistakes are concentrated more on plant pixels than on soil pixels. This suggests that the main difficulty is not general class confusion, but more specific ambiguity around plant boundaries and weak foreground structure (Fig. 4).

F. Ablation and Negative Results

The ablation experiments are useful because they show that not every plausible modification leads to improvement.

TABLE IV
DEEP-MODEL PERFORMANCE ACROSS TRAIN, VALIDATION, AND TEST SPLITS.

Model	Train Plant IoU	Val Plant IoU	Test Plant IoU	Test Plant F1	Test Soil IoU	Test Accuracy	Inference (s)
U-Net (Baseline)	0.7659	0.8317	0.8189	0.9005	0.9078	0.9349	1.0233
ResU-Net	0.8362	0.8424	0.8318	0.9082	**0.9157**	0.9405	1.1439
ASPP-ResU-Net	0.8390	0.8397	0.8321	0.9084	0.9155	0.9405	1.1977
AG-ASPP-ResU-Net	**0.8501**	**0.8447**	**0.8327**	**0.9087**	0.9156	**0.9406**	1.3510
Original paper best [2]	—	—	0.7750	0.8630	0.8990	0.9450	—

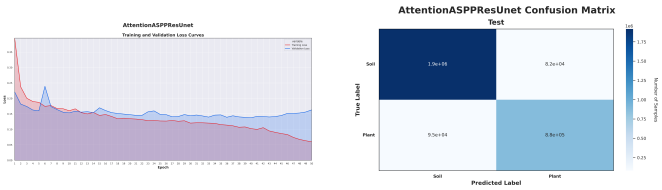
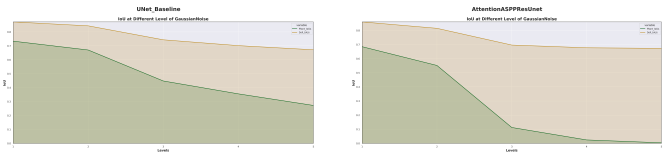


Fig. 4. AG-ASPP-ResU-Net. Left: training and validation loss curves showing stable convergence and a small late-epoch validation plateau. Right: test-set confusion matrix—residual errors are concentrated on plant pixels rather than soil, consistent with boundary-level ambiguity.



(a) U-Net (b) AG-ASPP-ResU-Net

Fig. 5. Ranking reversal under Gaussian noise. Plant/Soil IoU vs. severity. The plain U-Net degrades gradually (Plant IoU 0.735→0.271), while the residual-based AG-ASPP-ResU-Net collapses between levels 2 and 3 and reaches 0.005 at level 5, despite leading on clean data.

TABLE V
NEGATIVE-RESULT ABLATIONS.

Ablation	Main observation	Outcome
ExG concatenated into deep input	Slightly faster early convergence, but lower final quality than RGB-only baseline	Not adopted
Learnable residual scaling (α)	Slower optimization and lower final performance than standard residual block	Not adopted

Adding ExG as an extra deep-learning input sometimes makes early convergence slightly faster, but it does not improve final segmentation quality. This is an informative negative result, because ExG is highly relevant to vegetation segmentation and might have been expected to help. Instead, the result suggests that the convolutional models are already able to learn similar colour-ratio cues directly from RGB input.

A learnable residual-scaling variant also underperforms the standard residual design. With low alpha initialization, early gradients become weaker and convergence slows down, leading to lower final performance than the standard residual block.

The learnable-residual variant reached a test Plant IoU of about 0.8188, clearly below the standard ResU-Net result (0.8318). These negative results matter because they clarify what the final gains are *not* coming from. In both cases, adding more structure or more prior information did not help. This reinforces the idea that the strongest improvements in this task come from effective feature propagation rather than simply introducing more components.

G. Robustness Analysis

The robustness experiments reveal a fundamentally different picture from the clean-set ranking, and contain the most practically significant finding of the study.

Under affine rotation, performance remains very stable across all models, with almost no significant drop across levels. Under Gaussian blur, segmentation degrades moderately: for AG-ASPP-ResU-Net, Plant IoU falls from 0.831 to 0.791 at the highest severity, a decline that leaves the model still functional. JPEG compression shows a steeper trend, with Plant IoU dropping from 0.810 to 0.620, indicating that high-frequency boundary information matters more than it might appear. Brightness shift shows an asymmetric pattern: under-exposure causes substantially more degradation than overexposure, and the resulting curve is non-monotonic, suggesting that the models are more sensitive to the loss of visible structure than to moderate saturation.

Gaussian noise, however, produces a qualitatively different and more severe failure. At level 5 (variance 0.2), AG-ASPP-ResU-Net Plant IoU drops to 0.005—a near-complete collapse. More importantly, this vulnerability reveals a **ranking reversal** relative to clean-set performance. Between noise levels 2 and 3, ResU-Net Plant IoU falls sharply from 0.557 to 0.112; by noise level 5, all three residual-based models collapse to essentially zero Plant IoU (ResU-Net 0.003, ASPP-ResU-Net 0.005, AG-ASPP-ResU-Net 0.005). The plain U-Net, by contrast, degrades much more gracefully and retains a Plant IoU of 0.271 at the same severity. On highly noisy inputs, the clean-set ranking is therefore fully inverted: the model with the worst clean-set performance outperforms every model that surpassed it under ideal conditions (Fig. 5).

H. Failure Cases and Overall Findings

The failure cases bring the quantitative results into focus (Fig. 6). The hardest examples usually involve low-contrast plant boundaries, dark illumination, or severe image degradation. These are not isolated mistakes, but recurring patterns that match the corruption experiments and confusion-matrix trends.

TABLE VI
SUMMARY OF ROBUSTNESS BEHAVIOUR UNDER DIFFERENT CORRUPTIONS (AG-ASPP-RESU-NET; NOISE COMPARISON INCLUDES U-NET).

Corruption	Robustness	Key observation
Affine rotation	Very strong	Almost no drop across levels
Gaussian blur	Strong	Plant IoU 0.831→0.791 at highest severity
JPEG compression	Medium	Plant IoU 0.810→0.620; gradual boundary loss
Brightness shift	Uneven	Underexposure $\gg$ overexposure; non-monotonic
Gaussian noise	Weak	All residual models $\leq$0.005 at level 5; U-Net retains 0.271 (ranking reversed)

(a) Ground truth (b) Prediction

Fig. 6. Representative failure case (AG-ASPP-ResU-Net, test split). Thin low-contrast plant structures are under-segmented, producing fragmented predictions relative to the ground-truth mask.

In total, the results reveal a consistent pattern. Classical methods remain useful as efficient references, but their dependence on handcrafted features limits their ceiling. Deep learning models are substantially stronger, and all deep models exceed the original paper benchmark. Most of the meaningful clean-set gain comes from better optimization and feature propagation rather than from simply increasing complexity. Attention-Gated ASPP-ResU-Net gives the best clean-set result, but the robustness experiments reveal a serious weakness: the residual-based designs that improve clean-set learning are also associated with substantially weaker robustness under severe Gaussian noise, producing a ranking reversal that has direct implications for real-world deployment.

V. DISCUSSION

The results reveal a consistent pattern: the most reliable clean-set gains in this task come from improving feature propagation and optimization stability, rather than from increasing architectural complexity alone. This is most clearly reflected in the transition from U-Net to ResU-Net, which produces the largest single improvement among the deep models. By contrast, ASPP and attention gating still help, but mainly as refinements. Their contribution is real, yet much smaller than the gain introduced by residual learning. In this sense, the progression from U-Net to the strongest model is not simply a story of adding more modules, but of identifying which change addresses the main bottleneck most effectively.

The comparison with non-deep methods helps clarify that bottleneck. Among the traditional segmentation methods, the strongest results come from the simplest colour-based approaches: ExG (Plant IoU 0.754) and HSV thresholding (0.708). More elaborate alternatives such as DenseCRF, GrabCut, and Watershed all perform worse despite higher computational cost, which shows that procedural complexity does not automatically translate into better segmentation in a fully automatic setting. One clear reason is that the more complex methods still depend on the same colour prior (ExG) for initialisation; when that prior is imperfect, the refinement steps cannot fully recover. The failure of Watershed to achieve adequate Soil IoU—because insufficient soil-seed pixels are found in some images—illustrates this dependency directly.

ExG, HSV thresholding, Logistic Regression, and Random Forest all remain meaningful baselines because they are fast, interpretable, and grounded in useful colour information. At the same time, their performance clearly plateaus below that of the encoder-decoder models. This suggests that the main advantage of the deep models is not just better colour discrimination, but the ability to use spatial structure, local texture, and boundary information in a more coherent way.

The negative results reinforce this interpretation. ExG concatenation into the deep models is a particularly informative failure case: although ExG is a highly relevant vegetation prior, adding it as an extra input channel does not improve final deep-model performance. This suggests that the convolutional backbone already learns similar colour-ratio cues directly from RGB data, consistent with the finding in the original paper [2]. A similar pattern appears in the learnable residual-scaling variant, which adds parameterization but does not outperform the standard residual design. Together, these failed modifications indicate that the gains in this project do not come from adding more information or more flexibility by default, but from changes that improve how useful features are learned and propagated.

The robustness results add the most important qualification to the clean-set narrative. The ranking reversal under strong Gaussian noise is not a minor footnote: it reveals that the architectural change most responsible for clean-set improvement—the residual shortcut—is simultaneously associated with the greatest robustness failure. One plausible explanation is that the additive nature of residual connections allows noise injected at the input to propagate and accumulate across successive blocks, whereas the plain encoder of U-Net lacks this compounding path. GroupNorm, while effective for stabilizing training on clean data, may be misled by noise-corrupted activation statistics, potentially amplifying the problem at deeper layers. However, these are observations and candidate mechanisms, not confirmed causal claims: whether residual networks are categorically more or less noise-robust than plain networks is known to be task- and training-dependent, and targeted ablations would be required to isolate the contribution of each component. What the data do establish unambiguously is the ranking reversal itself, and its practical consequence: under severe Gaussian noise, architectural sophistication does not translate into robustness.

The validation-test gap is better understood as structural

than incidental. The 1.2 percentage-point drop in Plant IoU from validation to test reflects a year-based distribution shift—training and validation images were collected in 2018–2020, while test images are from 2017—and is therefore an expected consequence of temporal domain shift rather than a sign of unstable training.

In summary, the experiments suggest that the key challenge in this task is not the absence of architectural complexity, but the difficulty of learning representations that are both accurate on clean data and stable under realistic corruption. Further progress is unlikely to come simply from making the architecture more elaborate. More promising directions include training strategies that target noise sensitivity directly, such as noise-aware data augmentation, adversarial training, or architectural choices that decouple feature propagation from noise accumulation.

VI. Conclusion

The results show that deep encoder-decoder models are clearly more effective than classical pixel-wise baselines for plant-soil segmentation, with all deep models surpassing the original paper benchmark of Plant IoU 0.775. The clearest clean-set improvement comes from residual learning, while later additions such as ASPP and attention gating mainly act as smaller refinements.

The negative results are equally informative. Neither ExG concatenation nor learnable residual scaling improved the final deep-model performance, which shows that not every plausible enhancement is useful once the backbone is already strong. In that sense, the study does more than rank models: it helps clarify which kinds of modifications genuinely improve segmentation and which are largely redundant in this setting.

The main finding that goes beyond clean-set ranking is the robustness reversal under Gaussian noise: residual-based models collapse to near-zero Plant IoU at level 5 while the plain U-Net retains 0.271, exposing a concrete accuracy-robustness trade-off with direct implications for deployment in real agricultural environments. The study is additionally limited to binary segmentation. Future work should target training strategies that improve noise robustness without sacrificing residual-based clean-set gains, and extend the pipeline beyond binary plant-soil separation toward richer field understanding.

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